

Computational Financial Modelling Lab (MA-653)

Assignment: (4) Portfolio Optimization and Risk Management

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Dataset: IBEX35 Stocks (2010-2025)

1. Introduction

In this assignment we have used the stock data from the existing first two assignment, so data used here is clean, free from errors (as resolved earlier) and have no missing or NaN values.

In this, we have used the fundamental concept of Portfolio Optimization where we try to Minimize the Risk and Maximize the Return over a period of time.

Traditional portfolio theory focuses on Minimizing Variance(as we take variance as risk measure); however, modern risk management emphasizes tail-risk measures such as Value-at-Risk (VaR) and Conditional Value-at-Risk (CVaR)(and many more but we are using these only in this assignment).

This assignment implements a complete portfolio optimization pipeline using historical stock data from the IBEX35 index. The study explores:

- Global Minimum Variance (GMV) portfolio construction
- Risk estimation using VaR and CVaR
- Optimization under different risk objectives
- Cardinality-constrained portfolio selection
- Data-driven model comparison and recommendation

The analysis uses daily stock returns from 19 IBEX35 companies between 2010 and 2025 (as these are the companies which have maintained their position in the Index in last 15 years). All optimization models were implemented in Python(for simplicity) using numerical optimization techniques.

2. Data Description

The dataset consists of daily closing prices(in fact it has the data of *OHLCV*) for 19 stocks in the IBEX35 index covering a 15-year period. The data was cleaned and transformed before performing portfolio optimization.

Data Processing Steps

The following preprocessing steps were applied:

- Loaded historical stock price data from CSV files
- Selected closing price values
- Forward-filled missing observations
- Computed daily percentage returns
- Split dataset into training and testing sets

The training dataset contained:

- **19 stocks**
- **2672 trading days**

The training data was used to estimate portfolio weights and risk parameters.

3. Question-1 :- Global Minimum Variance Portfolio

3.1 Objective

The Global Minimum Variance (GMV) portfolio minimizes portfolio variance without imposing a return constraint.

The optimization problem is:

minimize $[\mathbf{w}^T \Sigma \mathbf{w}]$

subject to, $[\Sigma \mathbf{w}_i = 1]$ and $[\mathbf{w}_i \geq 0]$

3.2 Analytical Solution

The closed-form solution for the GMV portfolio is:

$$w(\text{GMV}) = (\Sigma^{-1} * \mathbf{1}) / (\mathbf{1}^T * \Sigma^{-1} * \mathbf{1})$$

Where:

- Σ = covariance matrix
 - $\mathbf{1}$ = vector of ones
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3.3 GMV Portfolio Performance

The computed GMV portfolio produced the following results:

Daily Return: **0.000545**

Annual Return: **14.72%**

Daily Volatility: **0.010804**

Annual Volatility: **17.15%**

These results indicate that the GMV portfolio achieves relatively low volatility while maintaining a stable return level.

3.4 Efficient Frontier

Figure: Q1EffFront.png

The efficient frontier represents the set of optimal portfolios that provide the maximum expected return for a given level of risk.

The GMV portfolio appears at the lowest point on the curve, indicating the minimum achievable volatility among all feasible portfolios.

3.5 Portfolio Weight Distribution

Figure: Q1GMVWs.png

The GMV portfolio allocated the highest weight to:

Red Eléctrica de España — **24.7%**

This allocation reflects the stock's low volatility and favorable correlation structure with other assets.

The result demonstrates a key principle of portfolio theory:

Assets with lower risk and lower correlation receive higher weight in minimum-variance portfolios.

4. Question-2 :- Risk Measurement using VaR and CVaR

4.1 Objective

Risk was evaluated using three different methods:

- 1. Historical Simulation
- 2. Parametric Normal Distribution
- 3. Parametric Student-t Distribution

The portfolio value used for risk reporting was:

INR. 10,000,000

4.2 Student-t Distribution Fit

The Student-t distribution was fitted to capture heavy tails in financial returns.

Estimated parameters:

Degrees of Freedom (v): **3.4561**

Mean: **-0.000563**

Scale: **0.009409**

A low degrees-of-freedom value indicates the presence of heavy tails, meaning extreme losses occur more frequently than predicted by a normal distribution.

4.3 Daily VaR and CVaR Results

Method	VaR95	VaR99	CVaR95	CVaR99
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Method	VaR95	VaR99	CVaR95	CVaR99
Historical	213,762	372,993	326,775	538,210
Normal	228,150	324,225	287,058	371,997
Student-t	204,378	379,638	323,038	556,131

Interpretation

The Student-t model produced the highest CVaR values, indicating greater tail risk compared to the normal model.

This confirms an important property of financial markets:

Returns exhibit fat-tailed behavior and cannot be accurately modeled using normal distributions alone.

4.4 Loss Distribution Visualization

Figure: Q2RiskThresholds.png

The histogram illustrates the distribution of portfolio losses along with VaR and CVaR thresholds.

The heavier tails visible in the distribution support the use of tail-risk measures such as CVaR.

5. Question-3 :- Portfolio Optimization Models

Three optimization models were implemented using different risk measures.

Model 1 – Minimum Variance

Objective:

minimize

$$\mathbf{w}^T \Sigma \mathbf{w}$$

This model focuses on reducing total portfolio volatility.

Model 2 – CVaR Minimization

Objective:

minimize

$\zeta + (1/(1 - \alpha)T) \sum u_t$

This model explicitly minimizes extreme losses and is widely used in risk management.

Model 3 – Semivariance Minimization

Objective:

minimize

$1/T \sum d_t^2$

This model penalizes downside risk only.

5.1 Portfolio Performance Comparison

Model	Return	Volatility	Sharpe	CVaR99
Equal Weight	0.094166	0.223794	0.152666	0.053821
Min Variance	0.135729	0.174924	0.432924	0.040280
CVaR	0.142687	0.182361	0.453426	0.038685
Semivariance	0.140302	0.175518	0.457517	0.039991

Interpretation

The CVaR model achieved the lowest tail risk, while the semivariance model produced the highest Sharpe ratio.

This demonstrates that downside-risk optimization methods provide better protection against extreme losses compared to variance-based methods.

5.2 Portfolio Weight Comparison

Figure: Q3GrpWeights.png

The grouped weight chart shows how different optimization objectives allocate capital across assets.

The CVaR model spreads risk more evenly across assets, reducing concentration in high-risk stocks.

6. Question-4 :- Cardinality-Constrained Portfolio Optimization

6.1 Objective

A cardinality constraint limits the number of assets in the portfolio.

Constraint: **$K = \text{ceil}(N / 3)$**

For this dataset:

$N = 19$

$K = 7$

This resulted in:

50,388 possible asset combinations

Each subset was evaluated to find the optimal sparse portfolio.

6.2 Performance Comparison

Variant	Return	Volatility	Sharpe	CVaR99
Model 1(Unconstrained)	0.135729	0.174926	0.432920	0.040280
Model 1(Cardinality)	0.135715	0.174926	0.432840	0.040281
Model 2(Unconstrained)	0.142687	0.182363	0.453420	0.038685
Model 2(Cardinality)	0.147170	0.185346	0.470312	0.038776
Model 3(Unconstrained)	0.140302	0.175519	0.457514	0.039991

Variant	Return	Volatility	Sharpe	CVaR99
Model 3 (Cardinality)	0.140346	0.175499	0.457812	0.039991

Interpretation

The cardinality constraint reduced diversification by limiting the number of assets.

However, in some cases, performance improved slightly due to the removal of weaker assets.

This illustrates a key trade-off:

Diversification vs simplicity.

7. Question-5 :- Data Analysis and Model Recommendation

7.1 Distribution Characteristics

The analysis revealed:

Number of stocks with fat tails: **19**

Positive skew: **11**

Negative skew: **8**

This indicates that extreme returns occur more frequently than predicted by normal distributions.

7.2 Sector Volatility Analysis

The most volatile sectors were:

Materials

Transport

Technology

The least volatile sector was:

This pattern reflects the defensive nature of utility companies compared to cyclical industries.

7.3 Model Ranking

The models were ranked based on:

- Expected return
- Volatility
- Sharpe ratio
- CVaR

Top-performing model:

Model 2 (budget + cardinality)

7.4 Recommended Model

Recommended portfolio:

Model 2 – CVaR Minimization (Unconstrained)

Justification

This model was selected because it:

- Achieved the lowest CVaR
- Provided strong Sharpe ratio
- Controlled extreme losses
- Maintained stable performance

CVaR is considered a coherent risk measure and is widely used in financial risk management.

8. Conclusion

This work demonstrated the application of portfolio optimization and risk management techniques using real financial data.

Key findings:

- Financial returns exhibit heavy-tailed distributions
- CVaR provides better protection against extreme losses
- Cardinality constraints improve practicality but may reduce diversification
- Risk-aware optimization leads to more stable portfolios

Overall, the CVaR minimization model provides the most reliable balance between return and risk and is recommended for practical portfolio management.

All the related Codes, Figures, CSV Docs can be found on the folder attached also on the GIT(<https://github.com/rajm012/Quant-Finance-Lab-Assignments/tree/main/Assignment-4>). For any issues or unclear explanation plz mail to b23406@students.iitmandi.ac.in or rajmahimaurya@gmail.com.

Logic, Idea and basic Snippets written by Raj but final code and report formatting has been done using the various LLM models (to look professional and reduce any grammatical errors).
